6 Fig. 6.1 shows an isolated conducting sphere which has been charged. Dashed lines (−−−−) join points of equal potential *V*. The potential difference between successive lines of equal potential is equal.

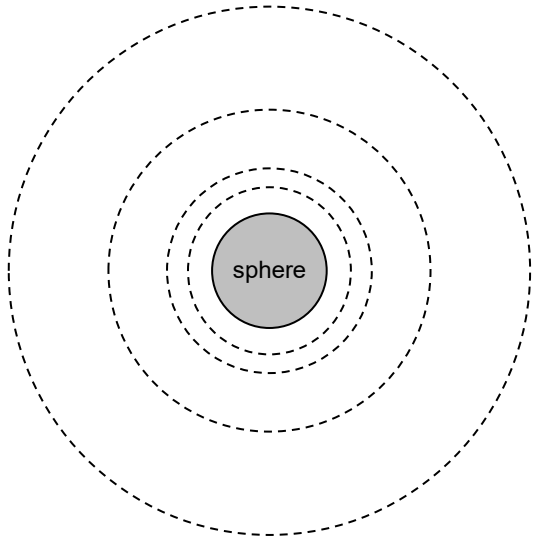

Fig. 6.1

For points on the surface or outside the sphere, the charge on the sphere behaves as if it were concentrated at the centre.

Measurements of the distance *x* from the centre of the sphere and the corresponding values of the potential *V* are given in Fig. 6.2. The values in Fig. 6.2 do not correspond to the dashed lines in Fig. 6.1.

x / m	*V* / V
0.19	-1.50×10^5
0.25	-1.14×10^5
0.32	-0.89×10^5
0.39	-0.73×10^5

Fig. 6.2

(a) On Fig. 6.1, draw the electric field lines. Label these lines *E*. [2]

(b) Explain how your drawing in **(a)** shows the relationship between electric potential *V* and the electric field *E*.

.....

.....

..... [2]

(c) **(i)** Use the data in Fig. 6.2 to show that the potential *V* is inversely proportional to the distance *x*. Explain your reasoning.

[2]

(ii) The potential at the surface of the sphere is -1.9×10^5 V. Calculate the radius of the sphere.

radius of sphere = ... m [2]

(iii) Determine the charge on the sphere.

charge = ... C [2]